R6.1: direct S_5 systems and the `chained-x` plug-in

Toy Koblitz curves only; no sub-rho claim; nothing about deployed curves. Every number below was measured by `examples/r6_1_direct_s5.rs` and the full tables are in `RESULTS.md`.

Verdict

Mixed: the falsifier is not met, and the hypothesis is partly refuted.

- **Falsifier not met.** No direct-S_5 arm beats the best m = 2 arm. At n = 13 the direct symmetrised S_5 costs S = 7.6e5 against 186 for mitm m = 2. From $\ell = 6$ ($n \geq 17$) the direct S_5 cannot be Weil-restricted inside a 15-minute cell.
- **Hypothesis partly refuted.** The direct S_5 is dearer per relation than the chain: 1.4e9 against 7.7e6 word XORs at n = 13. It is also not feasible wherever the chain is.
- **Discrepancy explained.** Matched target by target, the panel and the pipeline agree exactly: 396/396 outcomes and XOR counts. The 0.04 against 0.22 hit-rate gap is target sampling (Fisher p = 0.036), not a bug.
- **Plug-in.** The new `chained-x` oracle (library change) runs m = 3 and m = 4 end to end and recovers the planted logarithm. At every n the ordering is m = 2 < m = 3 < m = 4.

Interpolation

Both S_5 polynomials are unique, with F_2 coefficients:

- the symmetrised one has 182 unknowns, 24 orbits with coefficient 1 and Boolean degree 9;
- the x-frame one has 1 287 unknowns, 44 orbits and Boolean degree 12.

The same code reproduces the library's S_4 terms in both frames. Each polynomial vanished on 200/200 fresh relations at six (n, a) and on 0/200 random tuples.

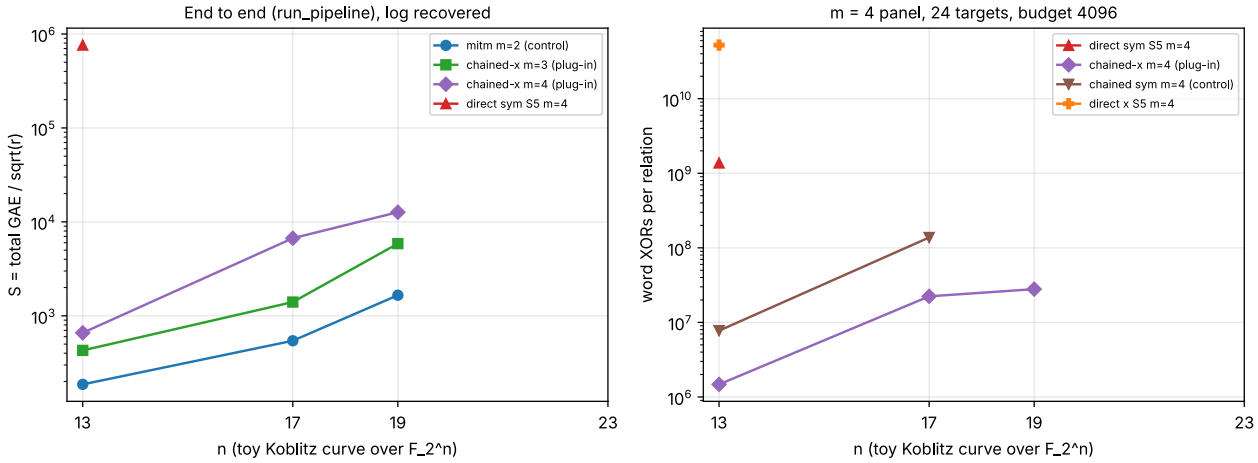


Figure 1: Left: S by arm and n from end-to-end `run_pipeline` runs, with the logarithm verified in every run. Right: word XORs per relation in the m = 4 panel, 24 targets per arm and 0 gate failures.

Limits

- n = 15 has no usable curve, and a = 1 exists only at n = 17.
- 6 cells were skipped and are marked NOT RUN.
- The symbolic build is not counted in XORs.
- All wall times are contended.